Guided Meta-Policy Search

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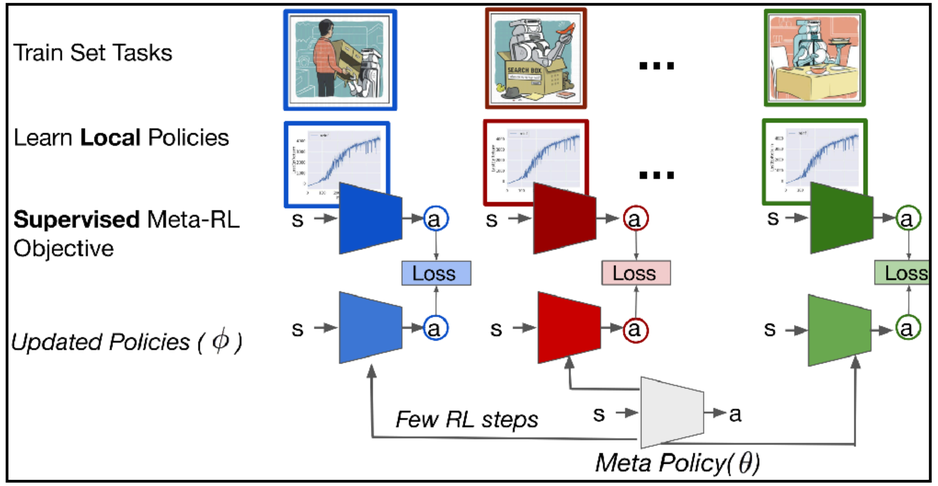
Motivation

- Training RL agents from scratch for each new task is very slow
- Meta-reinforcement learning enables effective use of prior experience for quick adaptation to new tasks.
- Current Meta-RL methods face challenges during meta-training:
  - extremely *high sample complexity*
  - Inability to train on harder tasks involving *sparse rewards* or *image observations*.

Decoupling Meta-Optimization

1. Train *local policies* for training tasks
2. Learned policies *guide* training of a meta-policy which can quickly adapt to perform any of the tasks.



- Sample efficiency gains due to *off-policy training*
- Optimization using *supervised objective* is more stable.

Method

Model Agnostic Meta-Learning

For each task i :

Test-time Adaptation : $\phi_i = \theta - \alpha \nabla_\theta \mathcal{L}_{\mathrm{RL}}(\theta, \mathcal{D}_i)$

Meta-train Objective : $\min_\theta \sum_i \mathcal{L}_{\mathrm{RL}}(\phi_i, \mathcal{D}_i)$

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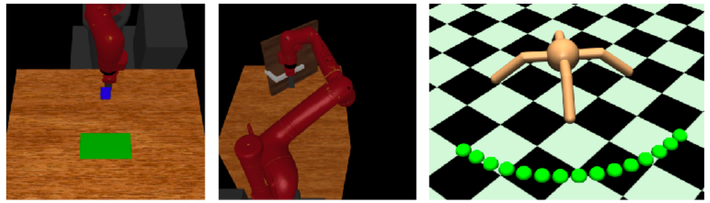
Importance Weighting

$$\phi_i = \theta + \alpha \mathbb{E}_{\tau \sim \pi_{\theta_{\text{init}}}} \left[\frac{\pi_\theta(\tau)}{\pi_{\theta_{\text{init}}}(\tau)} \nabla_\theta \log \pi_\theta(\tau) A_i(\tau) \right]$$

Meta-train Objective : $\min_\theta \sum_i \mathcal{L}_{\mathrm{BC}}(\phi_i, \mathcal{D}_i)$

$$\mathcal{L}_{\mathrm{BC}}(\phi_i, \mathcal{D}_i) \triangleq - \sum_{(\mathbf{s}_t, \mathbf{a}_t) \in \mathcal{D}_i} \log \pi_\phi(\mathbf{a}_t \mid \mathbf{s}_t)$$

Experiments



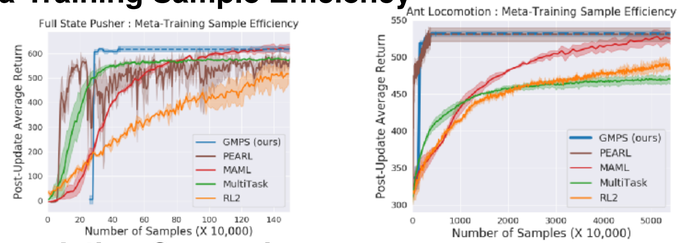
Sawyer Pushing: The arm needs to push the block to a goal position sampled from the green region.

Sawyer Door Opening: The door needs to be opened to a specific angle in the range from 0 to 60 degrees.

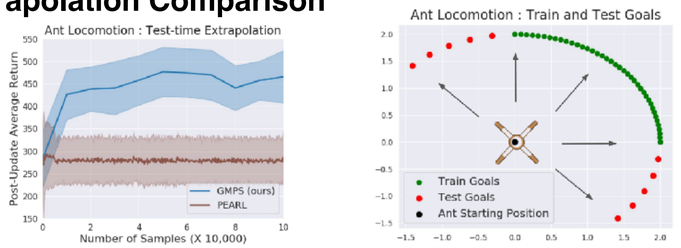
Legged Locomotion: The ant needs to walk to a goal point sampled from the quadrant shown.

Meta-Reinforcement Learning

Meta-Training Sample Efficiency

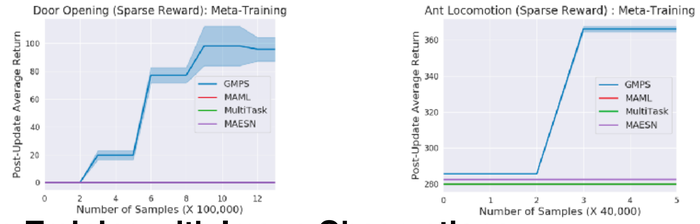


Extrapolation Comparison

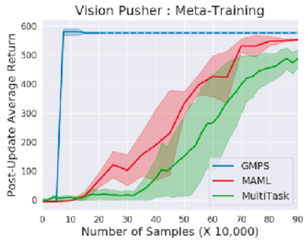


Meta-Learning from Demonstrations

Meta-Training with Sparse Rewards



Meta-Training with Image Observations



- Only GMPS detects reward for training on sparse reward tasks.
- GMPS is much more stable for vision-based policies